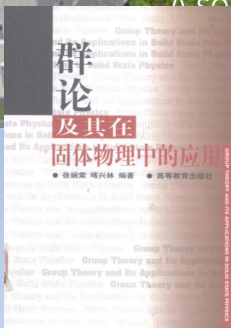


全国群论教学研讨会——第十八届年会



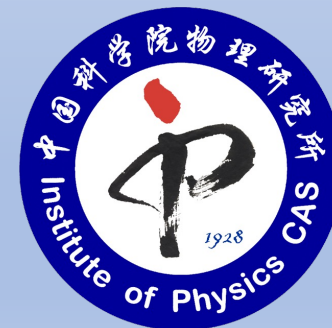
晶体能带表示理论和拓扑材料计算 TopMat介绍

<http://www.topmat.org>



王志俊

中国科学院物理研究所
凝聚态物质科学数据中心



2025年11月16日

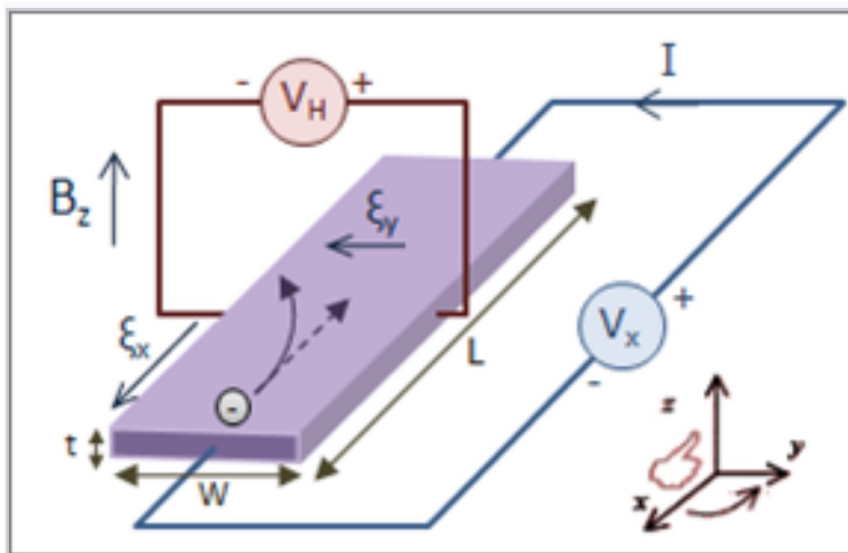
广东·深圳

目录

1. 拓扑物态和材料计算发现
2. 晶体能带表示理论（拓扑量子化学）
3. 能带表示的应用
 - a. 新型拓扑材料
 - b. 阻塞拓扑/非常规材料
4. 拓扑材料计算软件集（TopMat）

拓扑物态

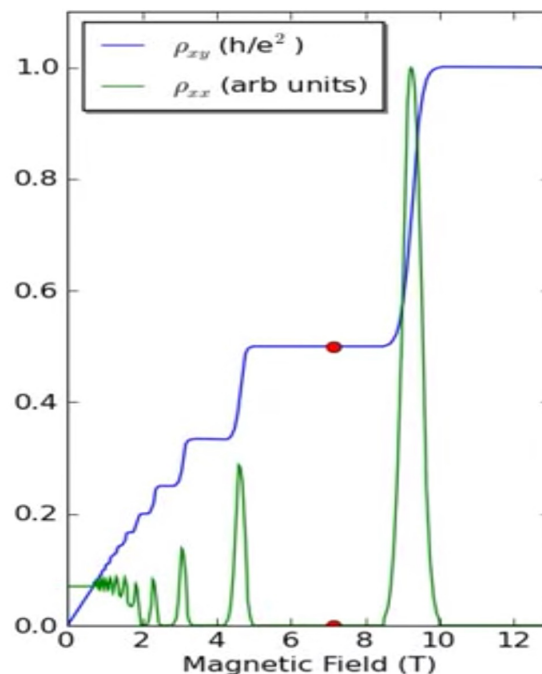
霍尔效应



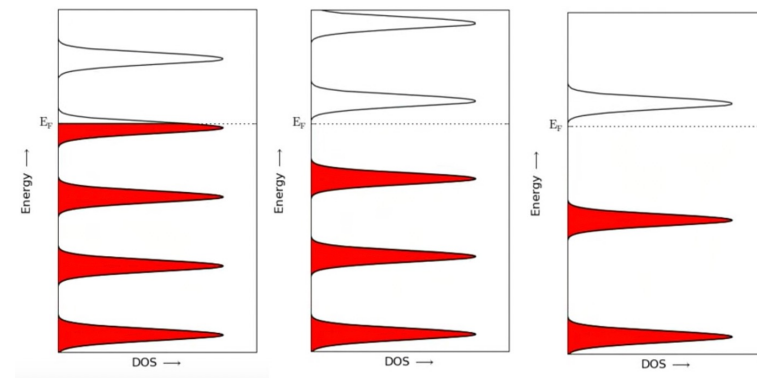
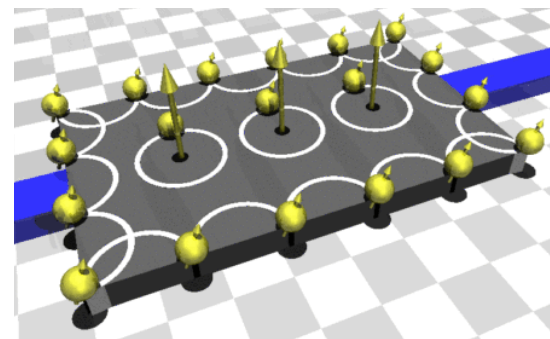
霍尔电阻 $R_{xy} = V_H / I_x = B / ne t$

量子霍尔效应 $R_{xy} = \frac{2\pi\hbar}{e^2} \cdot \frac{1}{i}, i \in \mathbb{Z}$

霍尔电导量子化~ TKNN number, 第一类Chern 数



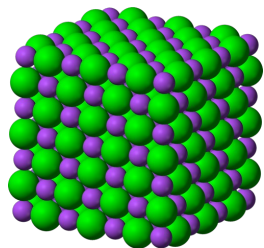
1. 宏观量子现象, 不随材料变化。
2. 拓扑稳定性, 不受杂质影响。



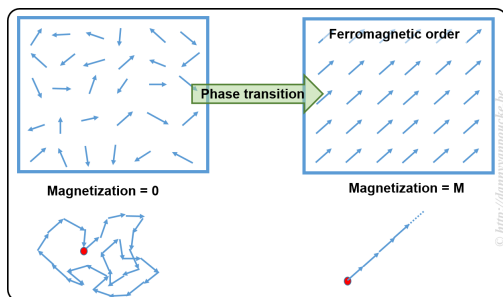
磁场B逐渐变大

拓扑物态

晶体中原子的有序排列
“破坏连续平移对称性”



磁铁中原子磁矩的有序排列
“破坏旋转对称性”



- **“拓扑物态”的发现，改变了人类对于“物态”的认知。这是全新的一大类物态，突破了朗道理论框架**
- **发现新型拓扑电子态可以扩展认知边界，成为当前物理学的重要科学目标**

新认知：物态包括两大类

“对称性物态”
朗道对称性破缺理论

+

“拓扑物态”
具有拓扑性质

拓扑 = 整数 = 量子

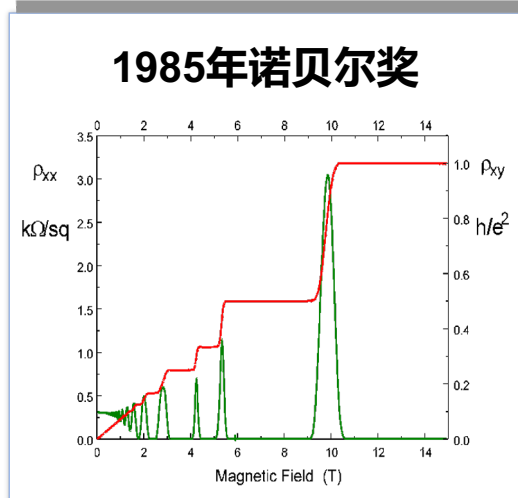
“对称性物态”只是世界的“一半”，另一半“拓扑物态”的大门刚刚打开



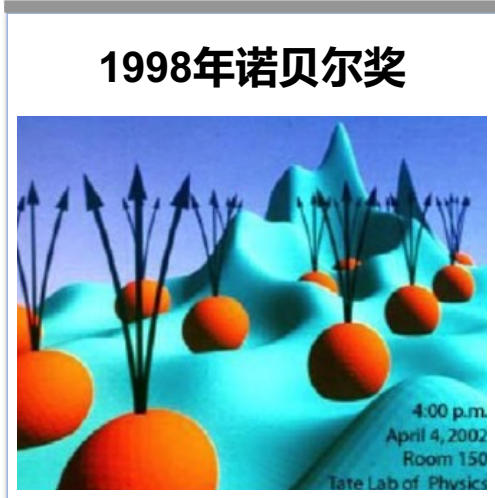
拓扑物态

- ▶ 近40余年来，该领域已**3**次获得诺贝尔物理奖
- ▶ 最近20年，又有了跨越式大发展，成为凝聚态物理的**重要新篇章**
 - 不同于前3次诺奖，在**固体材料**中发现了**全新拓扑态**
 - **计算预测**成为领域发展的强大驱动力（周期短，拓扑稳定，对称性）

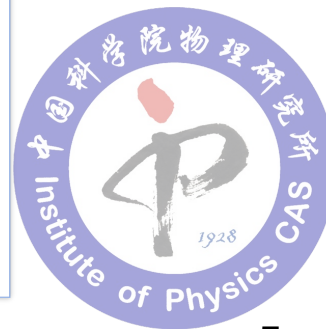
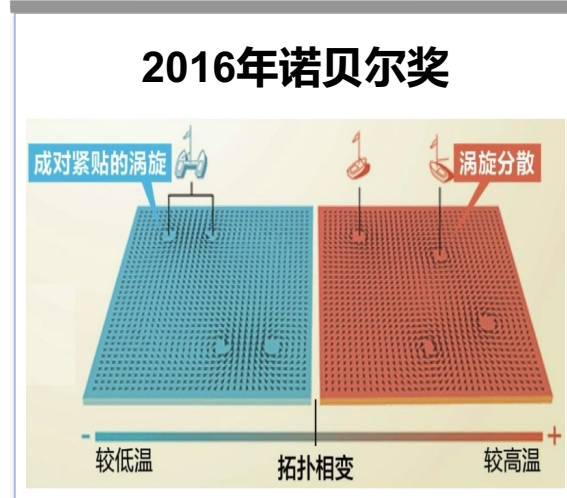
1980年
发现整数量子霍尔效应



1982年
发现分数量子霍尔效应

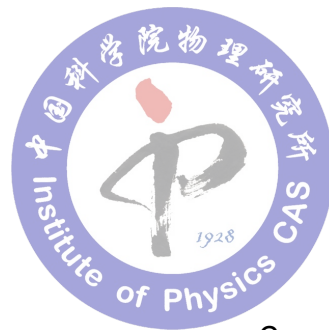


1973年
KT拓扑相变



拓扑材料的计算发现

我们组(HM)-T03的
在理论计算预言上
的做出了重要贡献。



Symmetry-protected topology

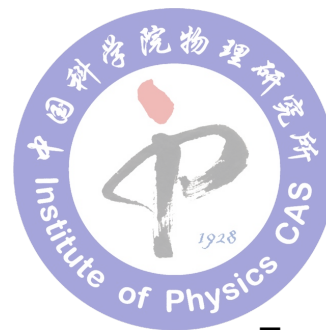
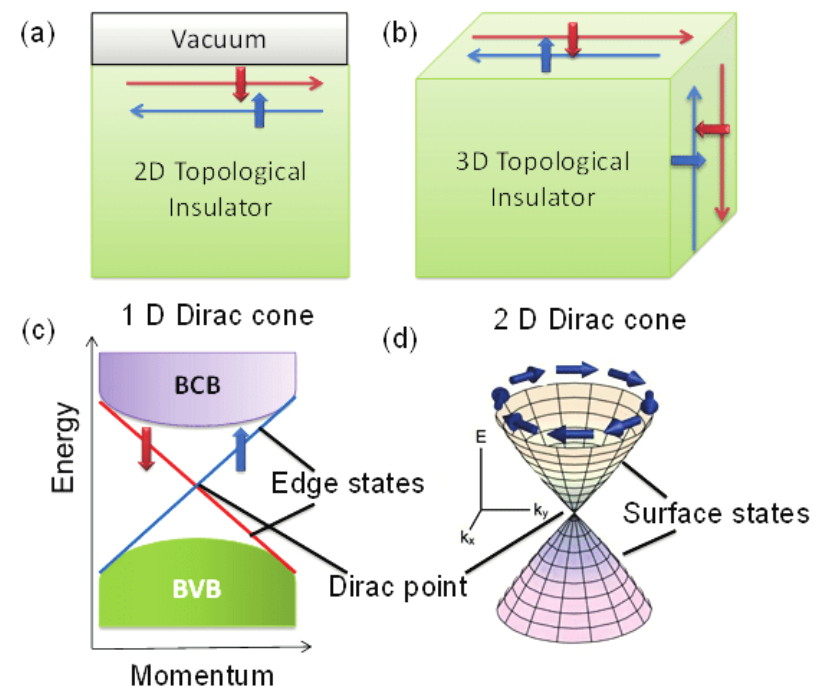
- Ten-fold Way
symmetry protected topological states (SPT)

T: time reversal C: charge conjugation

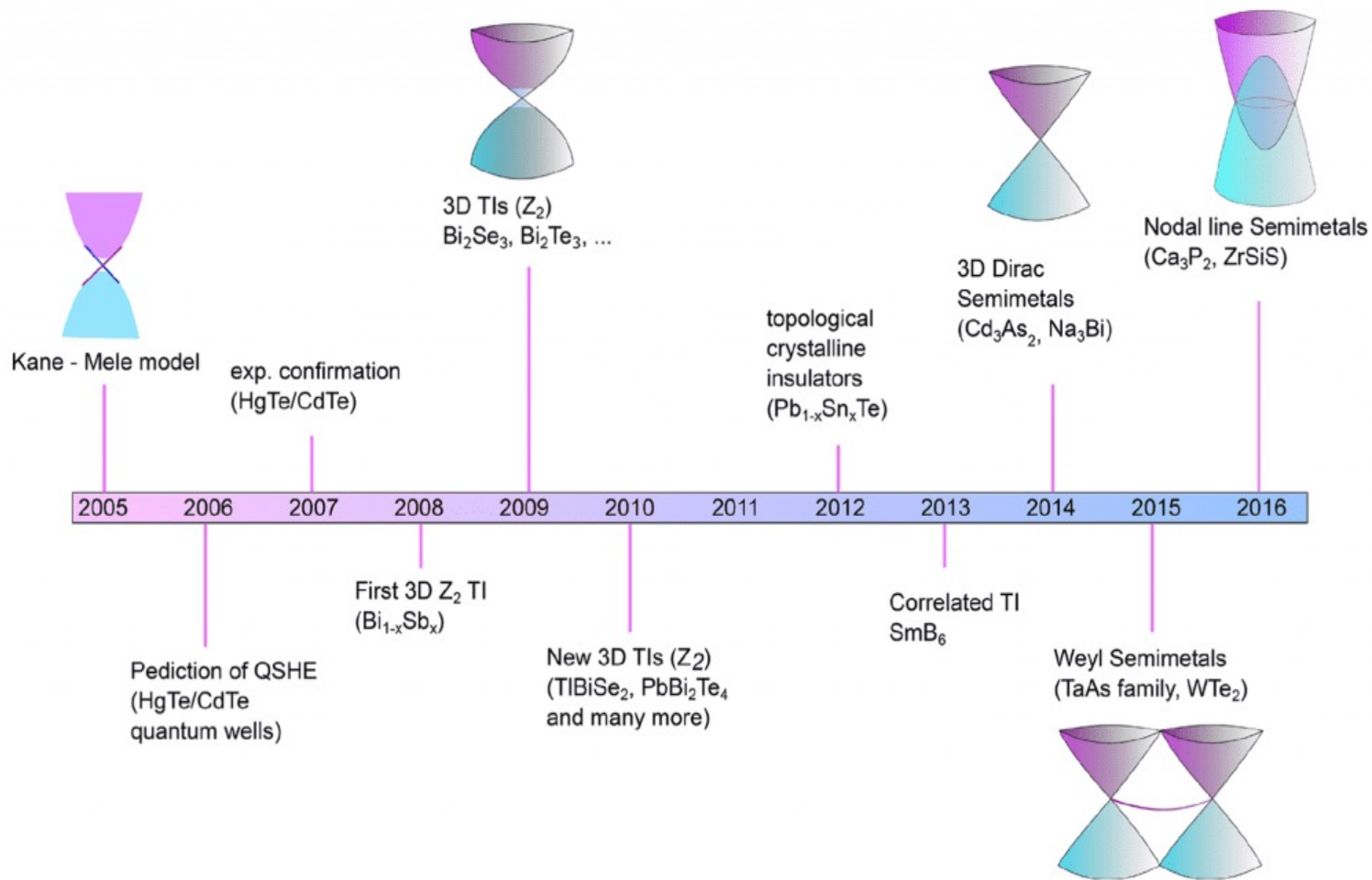
Class	T	C	S	0	δ							7
					1	2	3	4	5	6		
A	0	0	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	
AIII	0	0	1	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$	
AI	+	0	0	$\mathbb{Z}$	0	0	0	$2\mathbb{Z}$	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$	
BDI	+	+	1	$\mathbb{Z}_2$	$\mathbb{Z}$	0	0	0	$2\mathbb{Z}$	0	$\mathbb{Z}_2$	
D	0	+	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}$	0	0	0	$2\mathbb{Z}$	0	
DIII	−	+	1	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}$	0	0	0	$2\mathbb{Z}$	
AII	−	0	0	$2\mathbb{Z}$	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}$	0	0	0	
CII	−	−	1	0	$2\mathbb{Z}$	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}$	0	0	
C	0	−	0	0	0	$2\mathbb{Z}$	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}$	0	
CI	+	−	1	0	0	0	$2\mathbb{Z}$	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}$	

- Time reversal invariant systems
(All class)
QSHI; 3D STI

$$Z_2 \text{ index} = \sum_{\Gamma_\alpha \in TRIM} (n_\alpha^+ - n_\alpha^-)/2 \bmod 2$$

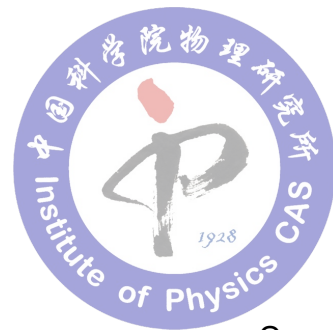


Topological materials in solids



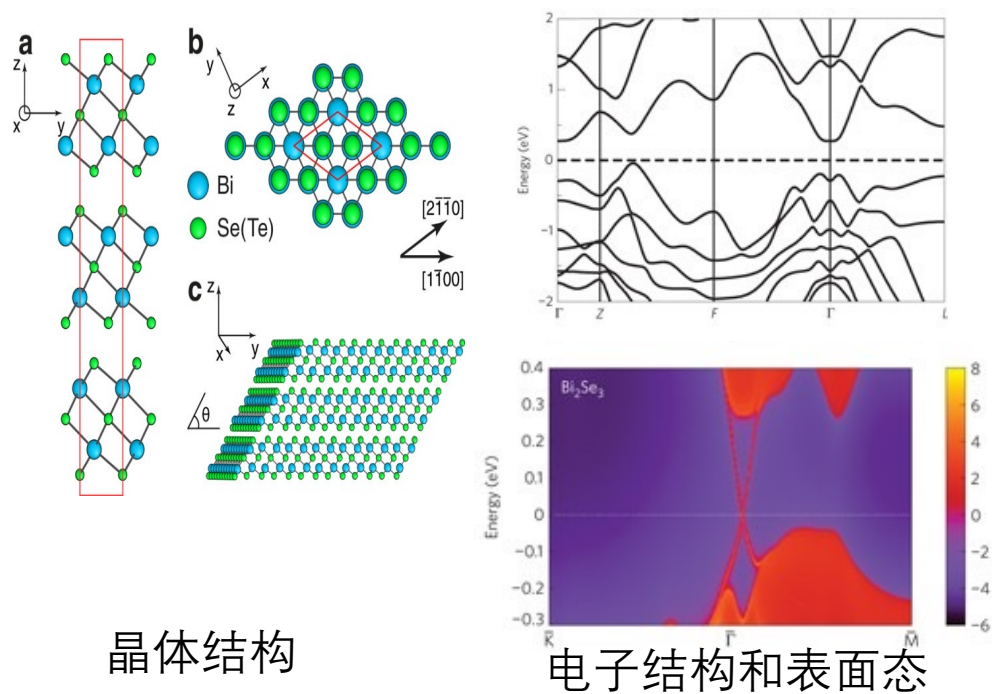
对称性保护的拓扑绝缘体、拓扑金属

- ✓ 3D TI (TR)
- ✓ Topological crystalline insulators (TCI) (mirror)
- ✓ 3D Dirac semimetals (C_3)
- ✓ Weyl semimetals (Translation)

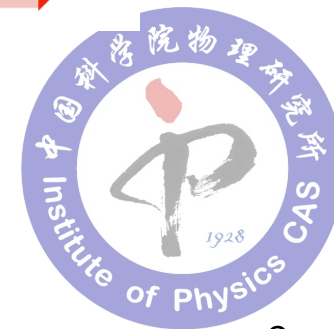
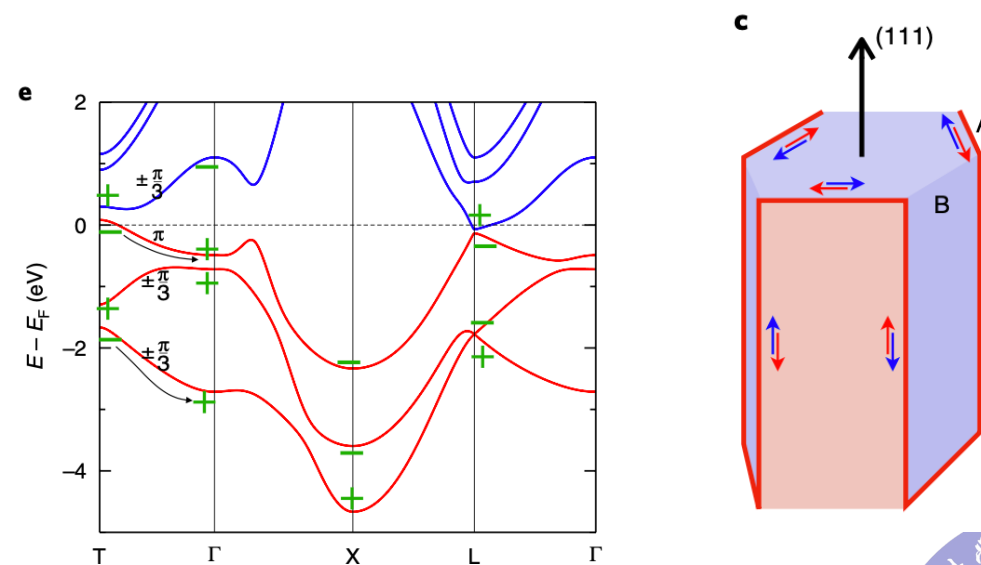


Topological materials

- Bi_2Se_3
3D STI (inversion Z_2)
Nat. Phys. 5, 438-442 (2009)



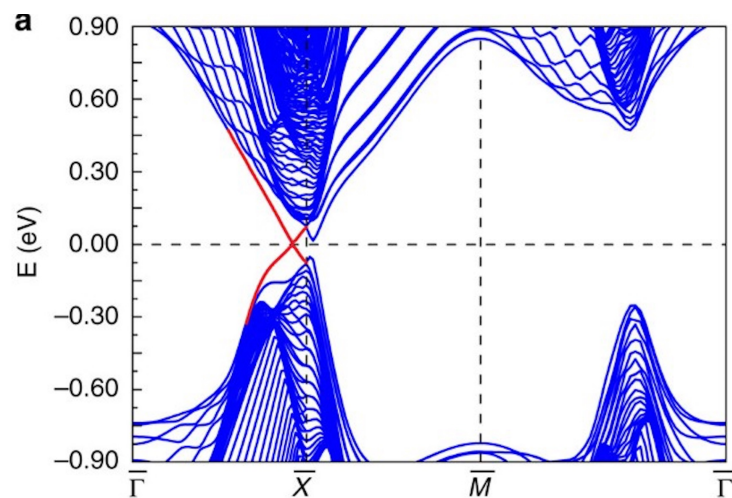
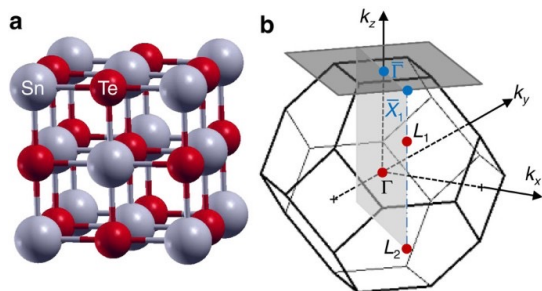
- Bismuth
higher-order TI (inversion Z_4)
Nature Physics 14, 918-924 (2018)



Topological materials

- SnTe

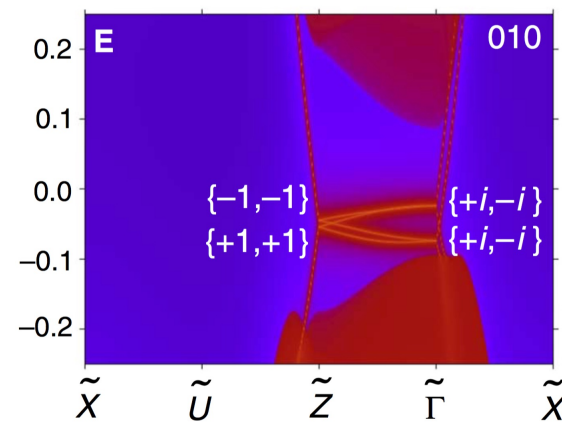
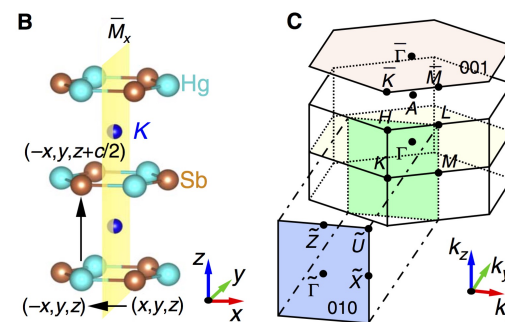
mirror symmetry protected TCI



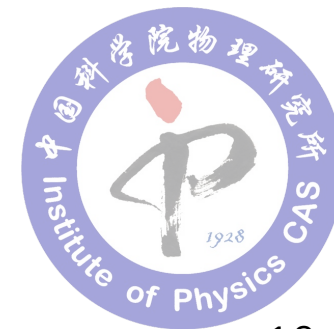
Hsieh, et al. Nat. Comm. 3, 982 (2012)

- KHgSb

glide symmetry (non-symmorphic TI)



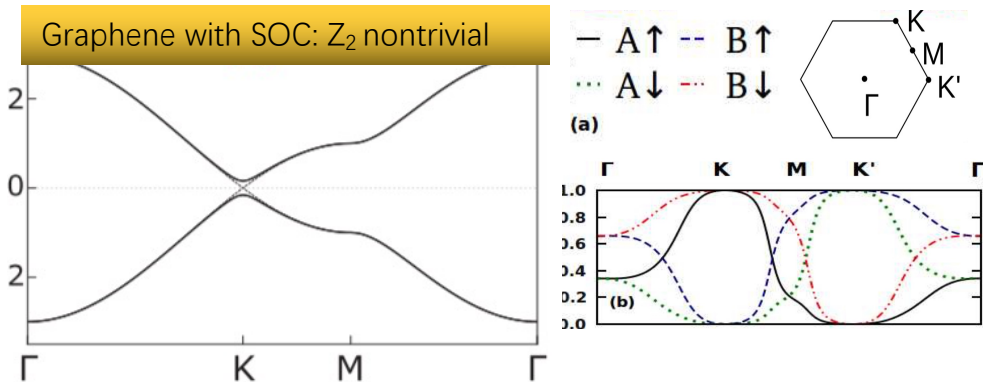
Wang, et al. Nature 532, 189 (2016)



Symmetry Indicators and Classification in 230 SGs

- Definition of TI/TCI in crystals
(No Wannier Representation; satisfying all the symmetries)

Graphene with SOC: Z_2 nontrivial



no time-reversal-invariant Wannier representation

Phys. Rev. B 83, 035108 (2011)

□ Topological quantum chemistry & band representations

Barry Bradlyn et al., Nature, 547 (2017)

1. Character tables for all k points in 230 SGs
2. Compatibility Relations
3. A full list of the elementary band representations (EBR)

▪ M.G., et al., Nature, 566, 480-485 (2019).

□ Symmetry-based classification & atomic orbitals

Po, H. C. et al., Nat. Comm. 8, 50 (2017)

1. Classifications

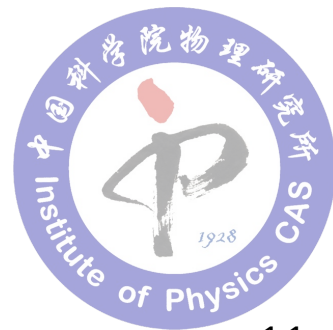
▪ Tang, F., et al., Nature 566, 486-489 (2019)

□ Layer construction & invariants

Z. Song et al., Nat. Comm. 9, 3530 (2018)

1. Fu-Kane-like symmetry indicators
2. Invariants for the TCIs

▪ Zhang, T., et al., Nature 566, 475-479 (2019)



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Topological quantum chemistry (TQC theory)

“Local orbital” in real space

Character Table of the group $C_{3v}(3m)^*$

$C_{3v}(3m)$	#	1	3	m_1-10	functions
Mult.	-	1	2	3	.
A_1	Γ_1	1	1	1	z, x^2+y^2, z^2
A_2	Γ_2	1	1	-1	J_z
E	Γ_3	2	-1	0	$(x,y), (xz,yz), (x^2-y^2, xy), (J_x, J_y)$

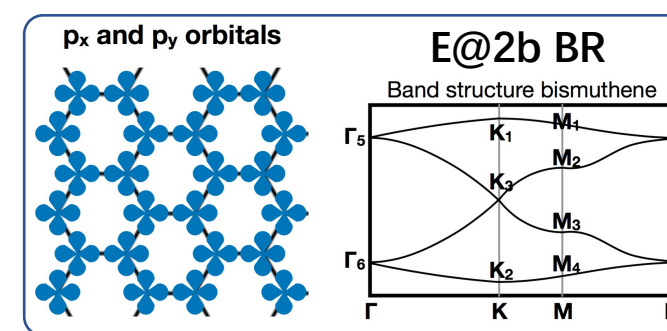
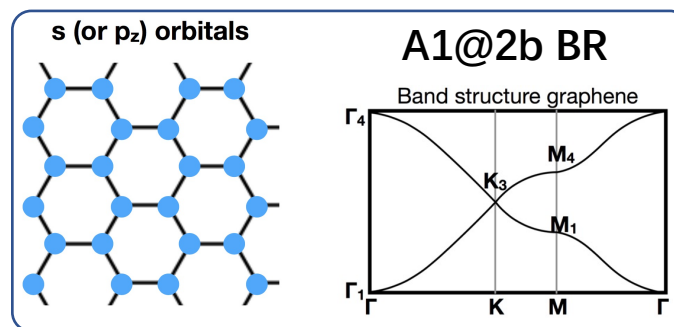
Site symmetry (2b Wyckoff site): $3m$

A_1 irrep : s or p_z

E irrep : p_x, p_y



“Band representation” in momentum space



TQC: Irreducible representations (IrRep), Compatibility Relations (CR), and Elementary band representations (EBR).



Published: 19 July 2017

Topological quantum chemistry

Barry Bradlyn, L. Elcoro, Jennifer Cano, M. G. Vergniory, Zhijun Wang, C. Felser, M. I. Aroyo & B. Andrei Bernevig

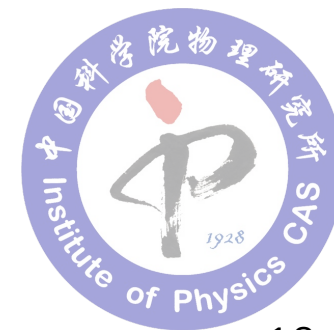
Nature **547**, 298–305 (2017) | [Cite this article](#)

76k Accesses | 755 Citations | 183 Altmetric | [Metrics](#)

□ 高通量计算

□ 材料拓扑指标计算

□ 非常规材料/OAI



拓扑量子化学的三个内容：

- Irreducible Representations (**IrRep**) for all k points
A band structure is represented by a set of IrRep.
-> vector data (including nonsymmorphic SGs)

(TQC basis)

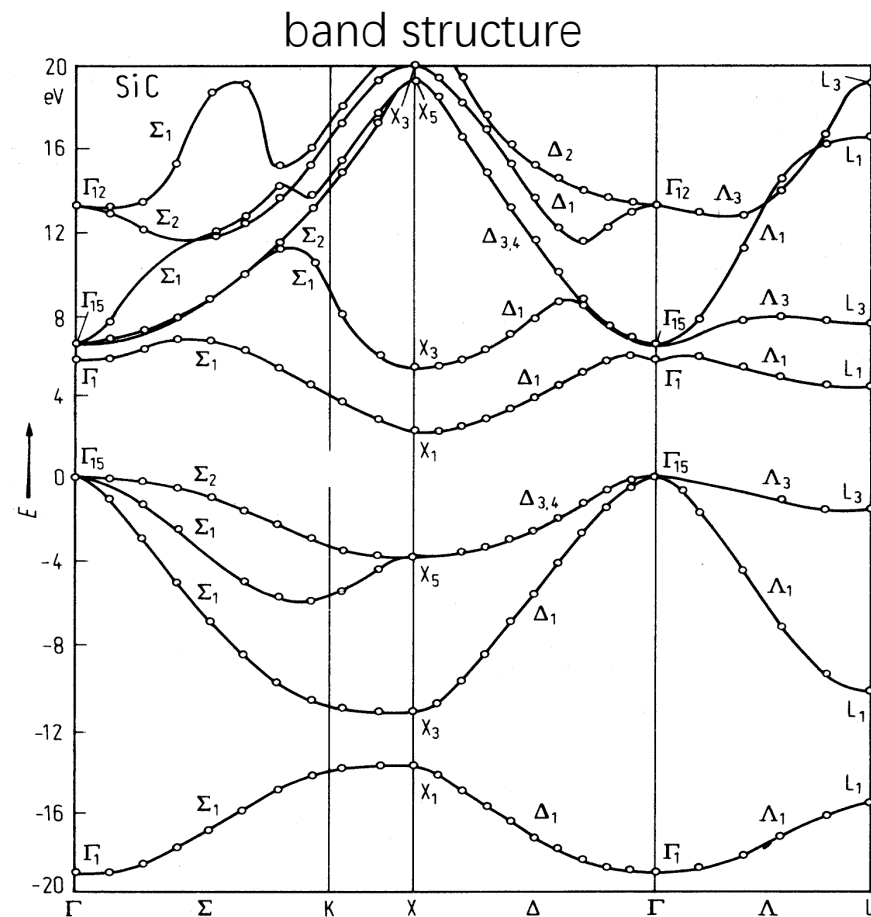
IRVSP: Gao, J. et al. *Comput. Phys. Comm.* 261, 107760 (2021)

IR2PW: Zhang, et al. *Phys. Rev. Research* 5, 023142 (2023)

- Compatibility Relations (**CR**, 相容性关系)
-> symmetry-based classifications and indicators.

- Elementary Band Representations (**EBR**, 基本能带表示)
All the trivial band structures are the sum of EBRs.
-> Topological materials?

能带表示分析



Ref: Bradlyn et al, NATURE 547, 298 (2017)

晶体空间群和能带表示理论

- Space group operator: $g = \{ R | t \}$
Space group G is a set of g operators.
- Translation group $T: \{ la_1 + ma_2 + na_3 \}$
 $\exp[-i k r]$, normal subgroup
- Symmorphic v.s. Non-symmorphic space groups
 $G = \{ T, g_2 T, g_3 T, \dots \}$
 $G = T \rtimes G_0$
Nonsymmorphic symmetry contains fractional translation.

- (Maximal) Wyckoff position (q)

Wyckoff Positions of Group $P6mm$ (No. 183)

Multiplicity	Wyckoff letter	Site symmetry	Coordinates
12	f	1	(x,y,z) $(-y,x-y,z)$ $(-x+y,-x,z)$ $(-x,-y,z)$ $(y,-x+y,z)$ $(x-y,x,z)$ $(-y,-x,z)$ $(-x+y,y,z)$ $(x,x-y,z)$ (y,x,z) $(x-y,-y,z)$ $(-x,-x+y,z)$
6	e	.m.	$(x,-x,z)$ $(x,2x,z)$ $(-2x,-x,z)$ $(-x,x,z)$ $(-x,-2x,z)$ $(2x,x,z)$
6	d	..m	$(x,0,z)$ $(0,x,z)$ $(-x,-x,z)$ $(-x,0,z)$ $(0,-x,z)$ (x,x,z)
3	c	2mm	$(1/2,0,z)$ $(0,1/2,z)$ $(1/2,1/2,z)$
2	b	3m.	$(1/3,2/3,z)$ $(2/3,1/3,z)$
1	a	6mm	$(0,0,z)$

Band Representation (BR; 能带表示)

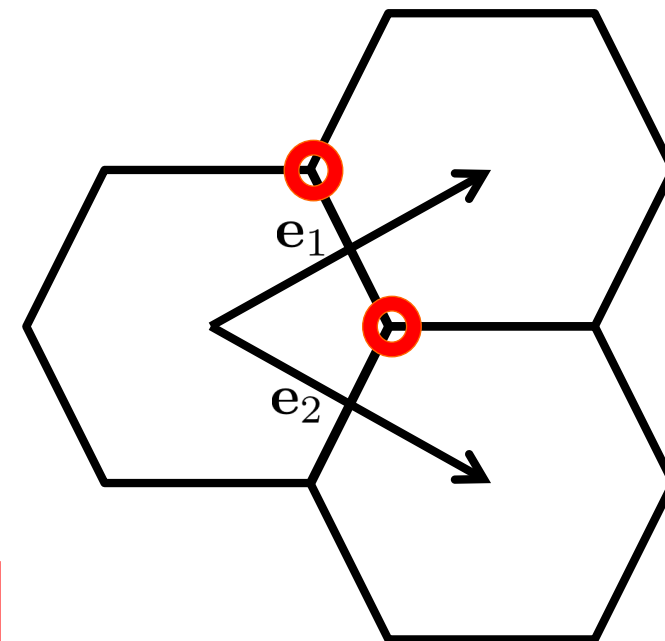
- Wyckoff position (multiplicity n ; $\{g_\alpha \mathbf{q}\}$)
- Representations (m-dimensional ρ -irrep)

$$a_{i\alpha}(\mathbf{k}, \mathbf{r}) = \sum_{\mu} e^{i\mathbf{k} \cdot \mathbf{t}_\mu} W_{i\alpha}(\mathbf{r} - \mathbf{t}_\mu),$$

- Matrix representations for all k-points.

$$(\rho_G(h)a)_{i\alpha}(\mathbf{k}, \mathbf{r}) = e^{-i(h\mathbf{k}) \cdot \mathbf{R}_{\beta\alpha}} \sum_{i'=1}^{n_{\mathbf{q}}} \rho_{i'i}(g_\beta^{-1}\{E| - \mathbf{R}_{\beta\alpha}\}hg_\alpha) a_{i'\beta}(h\mathbf{k}, \mathbf{r})$$

with $\mathbf{R}_{\beta\alpha} = hg_\alpha \mathbf{q} - g_\beta \mathbf{q}$



- Ref: Supplementary Information of “Topological quantum chemistry”, Nature 547, 298 (2017)



BR of 1a site

Wyckoff pos.	1a(6mm)	1a(6mm)	1a(6mm)	1a(6mm)	1a(6mm)	$\rho @ q$ BR	
Band-Rep.	$A_1 \uparrow G(1)$	$A_2 \uparrow G(1)$	$B_1 \uparrow G(1)$	$B_2 \uparrow G(1)$	$E_1 \uparrow G(2)$	$E_2 \uparrow G(2)$	$: q$
Decomposable\Indecomposable	Indecomposable	Indecomposable	Indecomposable	Indecomposable	Indecomposable	Indecomposable	$: \rho$
A:(0,0,1/2)	$A_1(1)$	$A_2(1)$	$A_4(1)$	$A_3(1)$	$A_6(2)$	$A_5(2)$	
Γ :(0,0,0)	$\Gamma_1(1)$	$\Gamma_2(1)$	$\Gamma_4(1)$	$\Gamma_3(1)$	$\Gamma_6(2)$	$\Gamma_5(2)$	
H:(1/3,1/3,1/2)	$H_1(1)$	$H_2(1)$	$H_2(1)$	$H_1(1)$	$H_3(2)$	$H_3(2)$	
K:(1/3,1/3,0)	$K_1(1)$	$K_2(1)$	$K_2(1)$	$K_1(1)$	$K_3(2)$	$K_3(2)$	
L:(1/2,0,1/2)	$L_1(1)$	$L_2(1)$	$L_4(1)$	$L_3(1)$	$L_3(1) \oplus L_4(1)$	$L_1(1) \oplus L_2(1)$	
M:(1/2,0,0)	$M_1(1)$	$M_2(1)$	$M_4(1)$	$M_3(1)$	$M_3(1) \oplus M_4(1)$	$M_1(1) \oplus M_2(1)$	

相容性关系只与空间群相关。

Ref: Bradlyn et al, NATURE 547, 298 (2017)

Classifications and SI

对于某个空间群，我们可以将所有最大高对称 k 点的不可约小表示可能出现的个数作为分量来构建矢量 p ,

$$p = (n(\Gamma_1), n(\Gamma_2), \dots, n(K_1), n(K_2), \dots)^T \quad (1.94)$$

$n(\Gamma_i)$ 为 Γ_i 表示在所关心的能带（一般为占据态）中出现的次数，在这里 n 的取

对 230 个空间群的分析表明，{CR} 矢量空间和 {BR} 的矢量空间的生成元的个数是相等的，因此商群一定为有限群。

$$X = \{CR\} / \{BR\} = \mathbb{Z}_{s_1} \times \mathbb{Z}_{s_2} \times \dots \times \mathbb{Z}_{s_N}$$

拓扑分类

任意一个 $m \times n$ 的整数矩阵 $\mathcal{B}$ ，我们都可以进行如下形式的史密斯分解，

$$L \cdot \mathcal{B} \cdot R = \Lambda \quad (3.19)$$

其中 L, R 分别为 $m \times m$ 维的行变换矩阵和 $n \times n$ 维的列变换矩阵，作用在 $\mathcal{B}$ 的左右两边可以对 $\mathcal{B}$ 进行初等行变换和初等列变换，它们都是可逆的。 Λ 为 $m \times n$ 维的对角矩阵，其对角元前 k 个元素为正整数，且从小到大排列，记为 λ_i ，其余元素均为零，

$$\Lambda = \begin{pmatrix} \Lambda' & \mathbb{0}_{k \times (n-k)} \\ \mathbb{0}_{(m-k) \times k} & \mathbb{0}_{(m-k) \times (n-k)} \end{pmatrix} \quad (3.20)$$

Λ' 为 Λ 中非零部分，为 $k \times k$ 的对角矩阵。

$$\Lambda' = \text{Diag}\{1, 1, 1, 1, 2\} \quad \begin{pmatrix} L^t \\ L^d \end{pmatrix} \cdot ((L^{-1})^l \quad (L^{-1})^r) = \begin{pmatrix} \mathbb{1}_{k \times k} & \mathbb{0} \\ \mathbb{0} & \mathbb{1}_{(m-k) \times (n-k)} \end{pmatrix}$$

$$\mathcal{B} \cdot R = L^{-1} \cdot \Lambda$$

$$(L^{-1})^l \Lambda' \cdot X = p$$

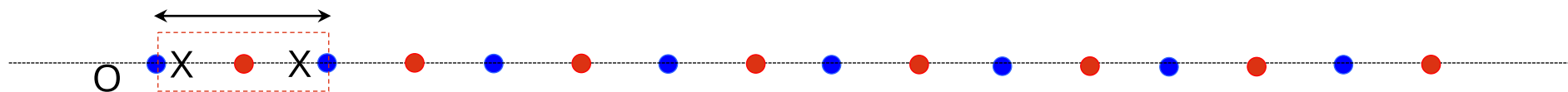
对称性指标： $\Lambda' \cdot X = L^t p$

EBR 对应的指标为 0； $(L^t)_i p \bmod \lambda_i = 0$

能带表示不是 EBR 的加和 等价于 拓扑指标非 0



EBR decomposition and RSI



EX: 1D inversion system ● A (0) ● B ($\frac{1}{2}$) x 2C



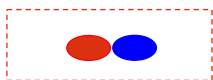
S@ A



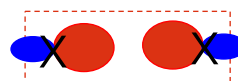
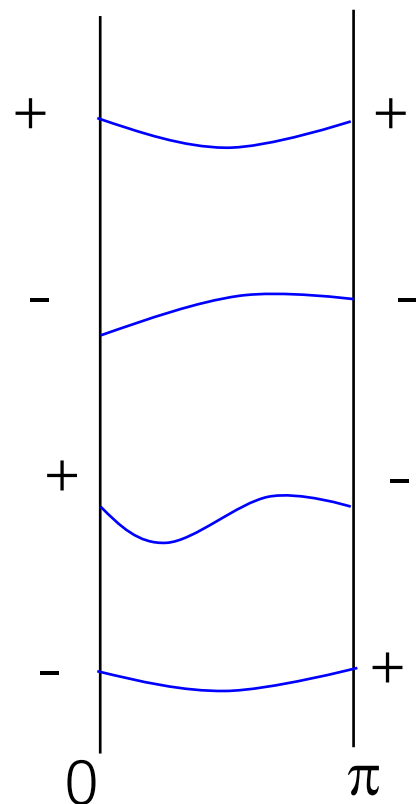
P@ A



S@ B



P@ B



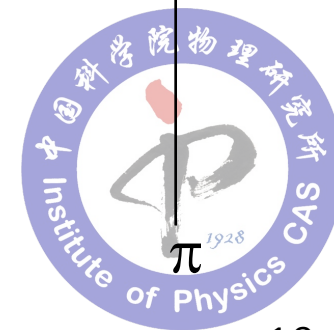
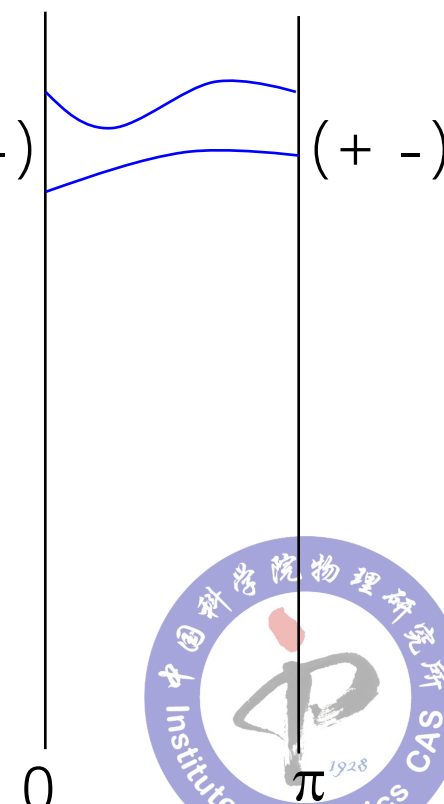
orb@ 2C

(+ -)

● Orb@2C 不能区分s,p

● orb@2C=(S+P)@A

● Real-space invariant(RSI)



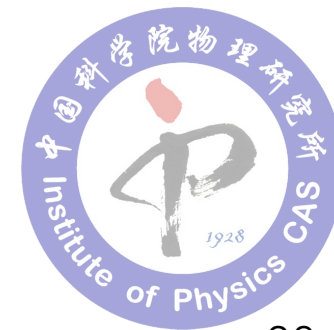
Search for topological materials by yourself

EX: Bi₂Se₃

- 1. The SG number (166) and crystal structure.
(run “**Phonopy**” to make sure the POSCAR is given in a standard setting.)
- 2. Get the high-symmetry k points (HSKPs) for that SG (166).
(https://github.com/zjwang11/irvsp/tree/master/max_KPOINTS_VASP/KPOINTS_166.txt).
- 3. Run “**VASP**” , to obtain the eigenstates (WAVECAR) at the several HSKPs.
(check the symmetry operators (irot) in OUTCAR)
For VASP6.x, please modify the line "CALL SET_SPINROT_WRAPPER(LATT_CUR%B(1,1),-1)" in mkpoints.F
- 4. Run “**IRVSP**” that you have just installed locally.
(https://github.com/zjwang11/irvsp/blob/master/src_irvsp_v2.tar.gz
tqc.data would be generated automatically)
- 5. Upload “tqc.data” and press the button. (<http://www.topmat.org> -> TopMat and MOM2MSG)

Ref: Gao, J. et al. [Phys. Rev. B 106, 035150 \(2022\)](#)

IU6

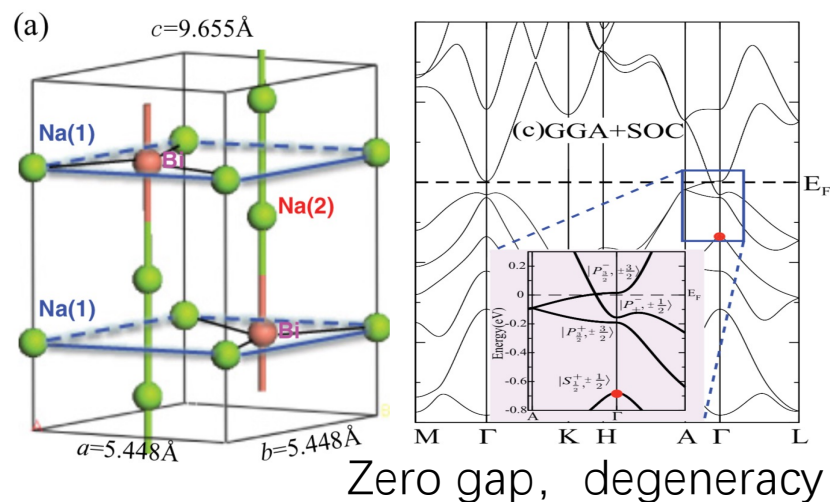


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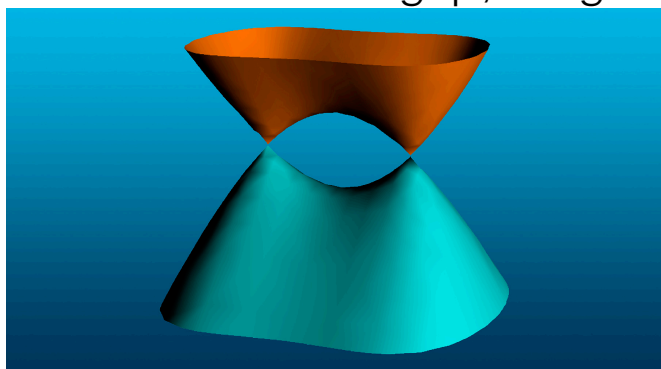
1. 拓扑物态和材料计算发现
2. 晶体能带表示理论（拓扑量子化学）
3. 能带表示的应用
 - a. 新型拓扑材料
 - b. 阻塞拓扑/非常规材料
4. 拓扑材料计算软件集（TopMat）

新型拓扑半金属的发现

3D Dirac semimetal (Na₃Bi)



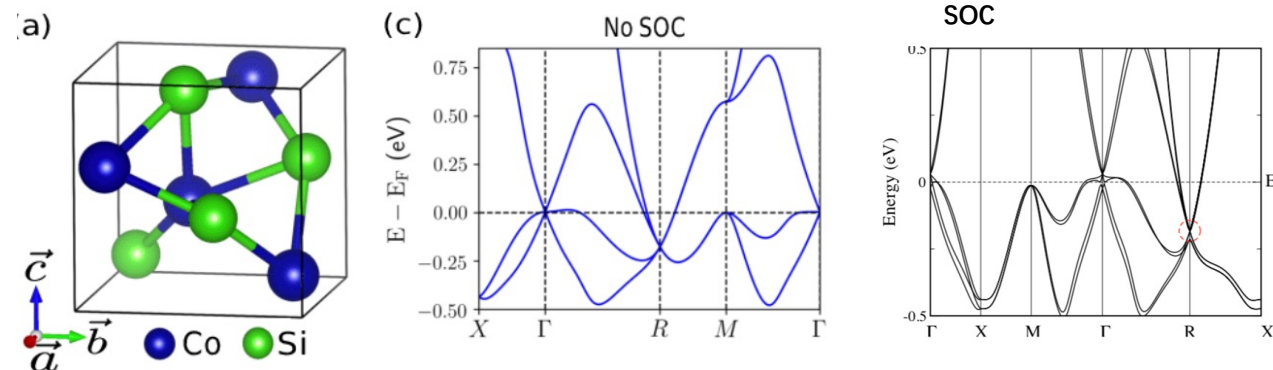
Zero gap, degeneracy



PRB 85, 195320 (2012)

Threefold (spin-1) particle (CoSi)

$$H_{199}(\phi, \delta \mathbf{k}) = \begin{pmatrix} 0 & e^{i\phi} \delta k_x & e^{-i\phi} \delta k_y \\ e^{-i\phi} \delta k_x & 0 & e^{i\phi} \delta k_z \\ e^{i\phi} \delta k_y & e^{-i\phi} \delta k_z & 0 \end{pmatrix} \quad (1)$$

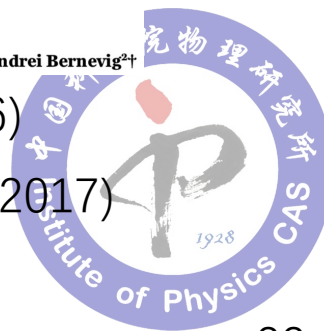


Beyond Dirac and Weyl fermions: Unconventional quasiparticles in conventional crystals

Barry Bradlyn,^{1*} Jennifer Cano,^{1*} Zhijun Wang,^{2*} M. G. Vergniory,³ C. Felser,⁴ R. J. Cava,⁵ B. Andrei Bernevig^{2†}

Science, 10.1126/science.aaf5037 (2016)

PRL, 10.1103/PhysRevLett.119.206402 (2017)



新型简并准粒子的发现

新型拓扑半金属（230空间群的低能有效模型）-没有高能粒子对应的准粒子

二度简并准粒子（Weyl）

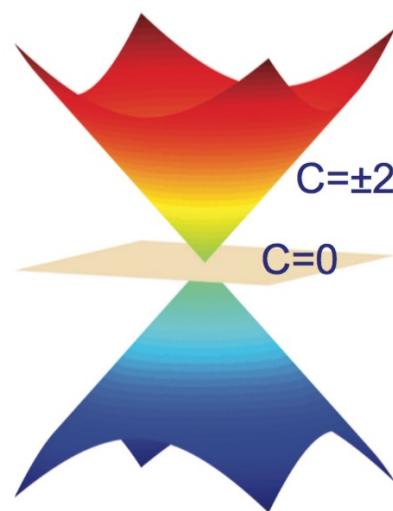
(a) Spin-1/2 Weyl fermion



$$H_W = k \cdot \sigma$$

三度简并准粒子

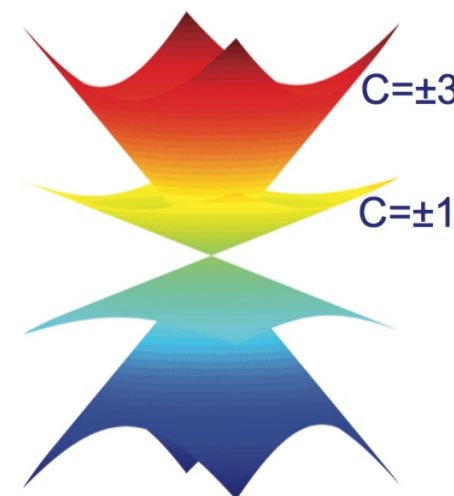
(b) Spin-1 excitation



$$H = k \cdot D^{(1)}$$

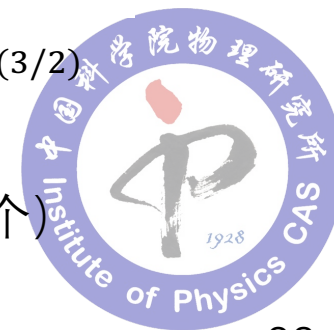
四度简并准粒子

(c) Spin-3/2 RSW fermion

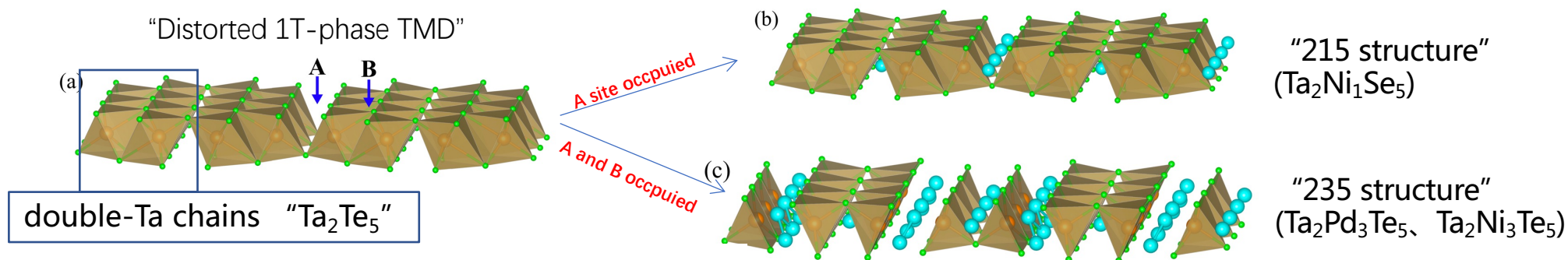


$$H = k \cdot D^{(3/2)}$$

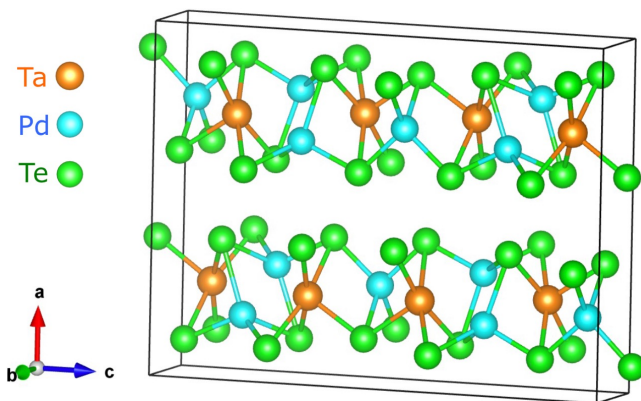
$D^{(1/2)}, D^{(1)}, D^{(3/2)}$ 三维转动旋转群SO (3)在不同基矢下 ($l=1/2, 1, 3/2$) 的生成元 (只有三个)



Higher order TI $\text{Ta}_2\text{M}_3\text{Te}_5$ (M=Pd, Ni)

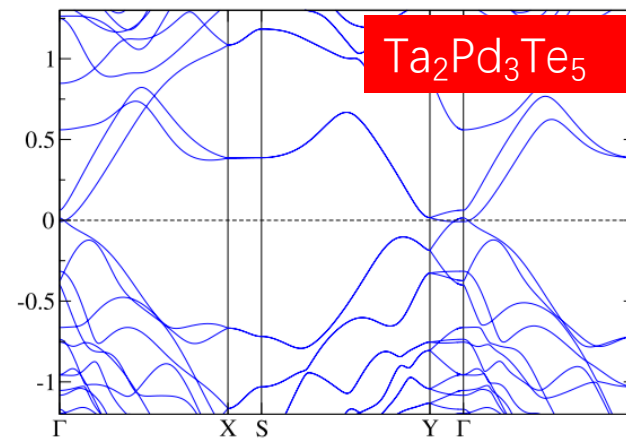
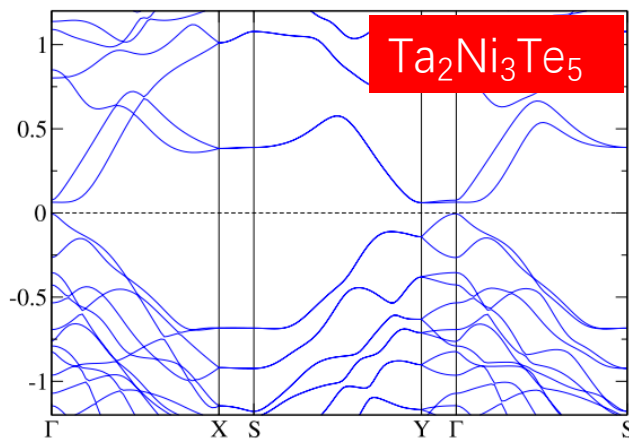


❑ vdW layers of the 3D bulk crystals



Two vdW layers in a unit cell
b is the 1D chain direction.

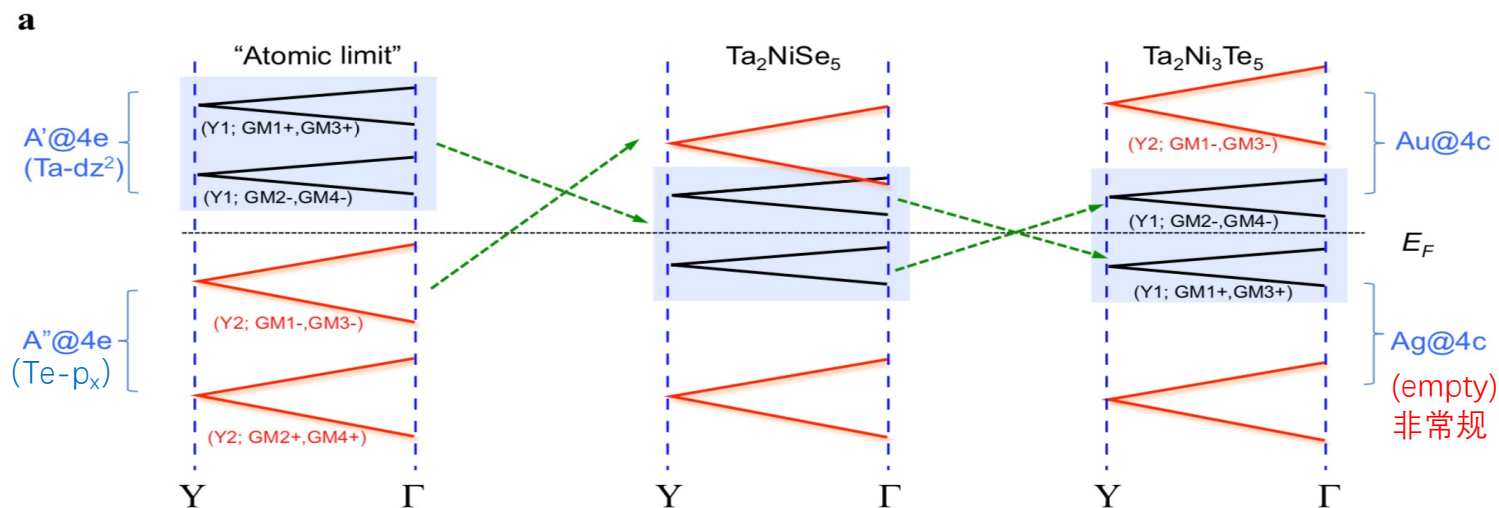
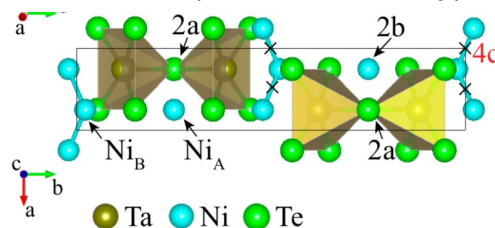
❑ Band structures of the 2D monolayers



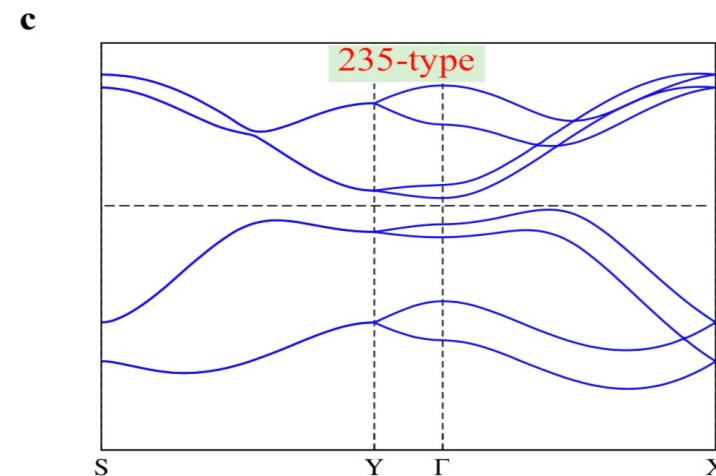
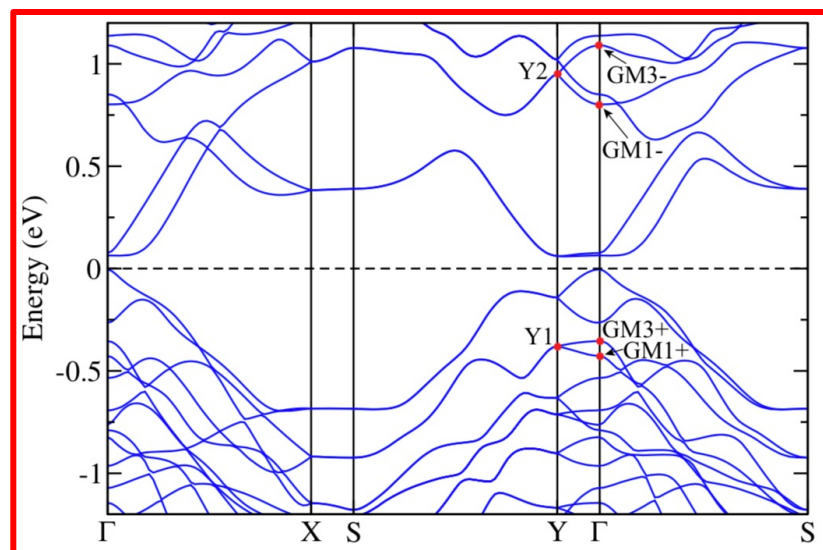
Double band inversion

atomic valence states :
 Ta^{5+} ; Te^{2-}

atomic positions in 235-type



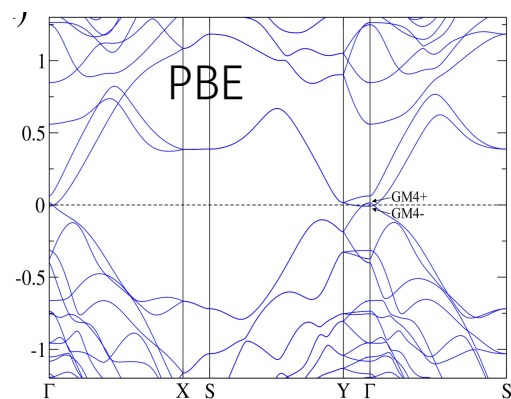
$\text{Ta}_2\text{Ni}_3\text{Te}_5$ 的电子能带



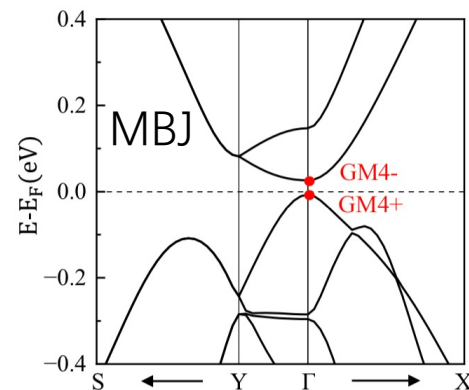
Higher-order topology
 ——2D Quadrupole TI

Excitonic insulator $\text{Ta}_2\text{Pd}_3\text{Te}_5$

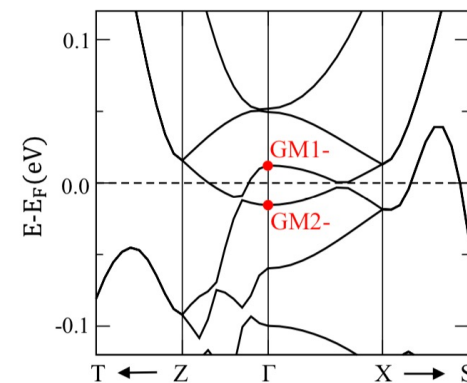
1. Small band gap.
2. Weak screening effect



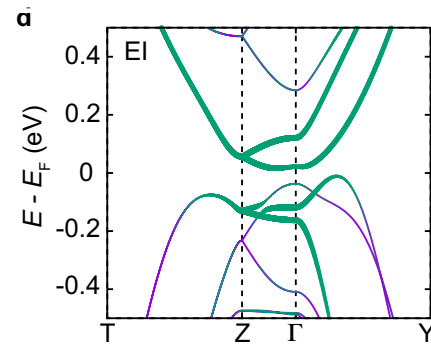
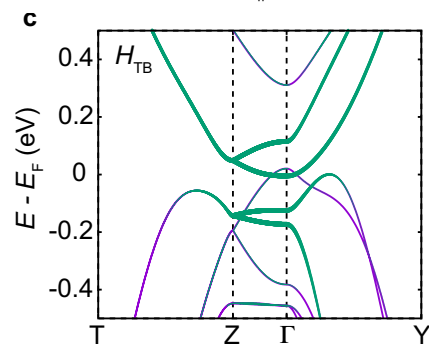
Band-edge states have opposite m_z symmetry



Band-edge states have the same parity in bulk

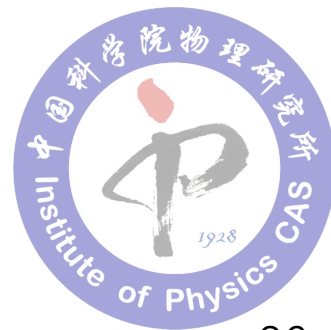
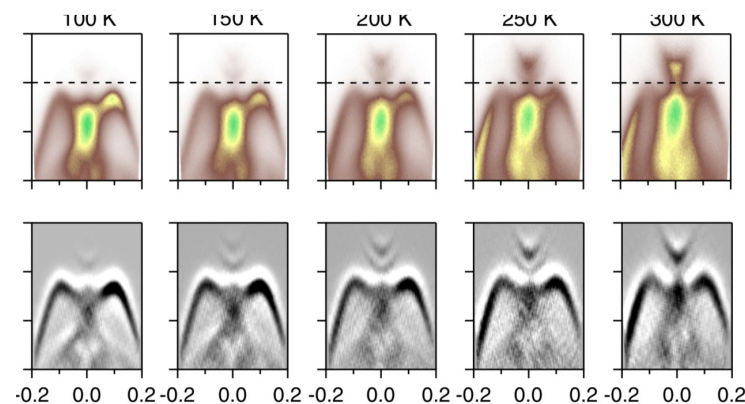


GW+BSE indicates it is an Excitonic Insulator



1. Chin. Phys. Lett. 41, 097101 (2024); Phys. Rev. B 103, 115145 (2021).
2. PHYS. REV. X 14, 011046 (2024); PRX 14, 011047 (2024)

激子态实验证据：激子能隙随着温度降低逐渐变大。

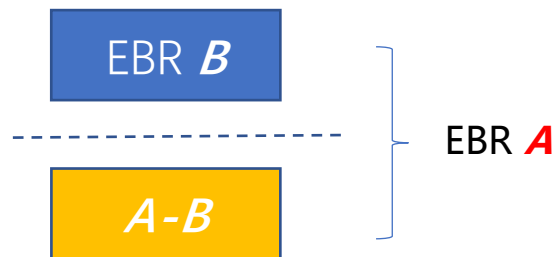


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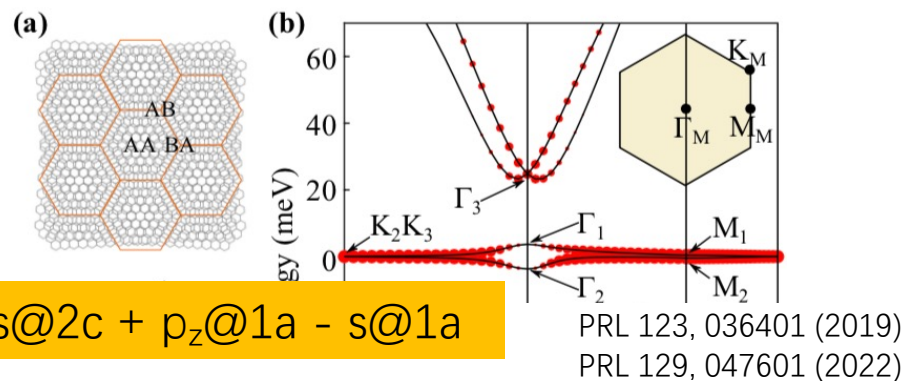
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Obstructed topology

□ Fragile topology



EX : Twisted Bilayer Graphene (TBG)

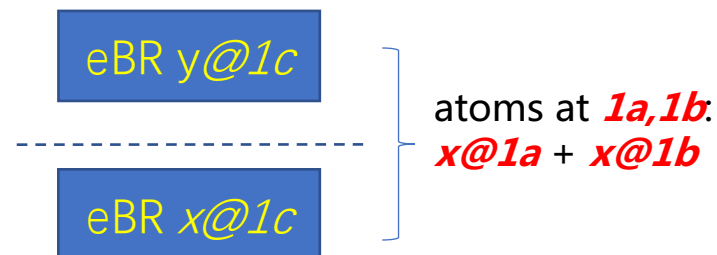


Topological Crystalline Insulators

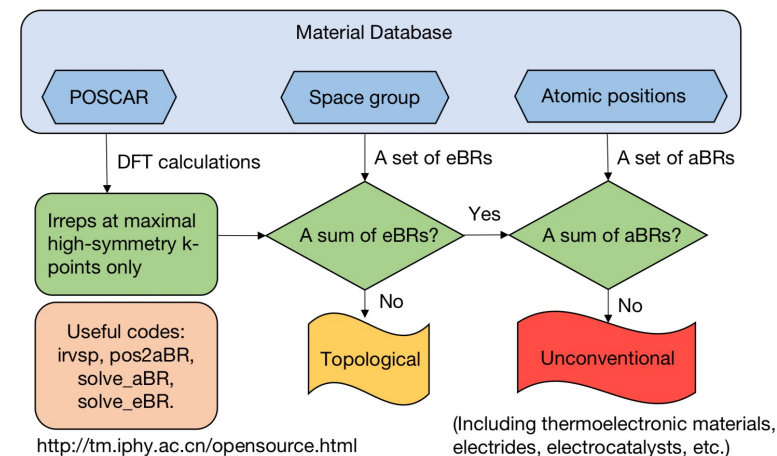
Liang Fu
Phys. Rev. Lett. **106**, 106802 – Published 8 March 2011

□ Obstructed topology

Unconventional nature/obstructed atomic limit

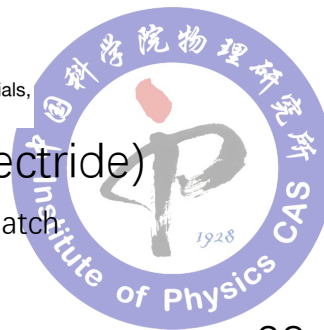


高的化学活性：热电，催化，储氢，超导等



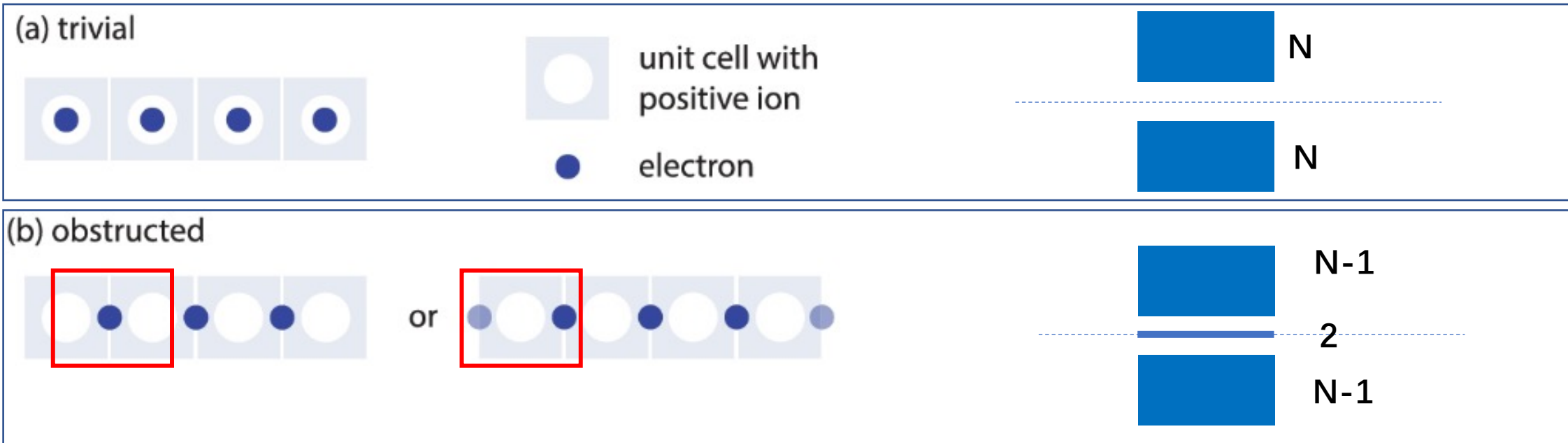
“free” electron in an interstitial site (Electride)

Gao, J. et al. "Unconventional materials: the mismatch between electronic charge centers and atomic positions", Sci. Bull. 67, 598 (2022)



Filling anomaly and RSI

Wannier Charge Centers do not sit on the atoms



filling anomaly :
Not satisfy both symmetry and charge neutrality

Diagnose unconv. Mat. (OAI)

1. Not a sum of ABR
2. Characterized by the Real space invariant (RSI)

W. Benalcazar et al. Phys. Rev. B 99, 245151 (2019)
Y. Xu et al. arXiv:2111.02433 (2021)

Classification

➤ Electronic charge centers (band representation decomposition in TQC)

□ Ordinary insulator

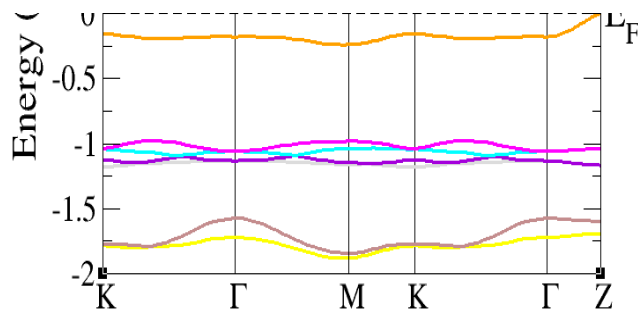
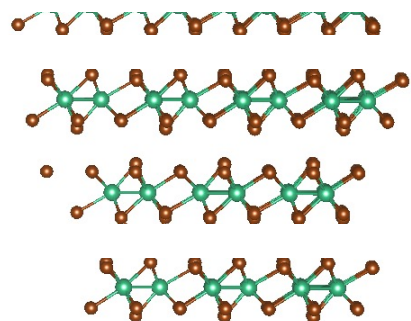
□ Topological insulator
(No Wannierizable)

□ Unconventional insulator
(Obstructed atomic limit)

alpha-Nb₃Br₈ (low-temp.)



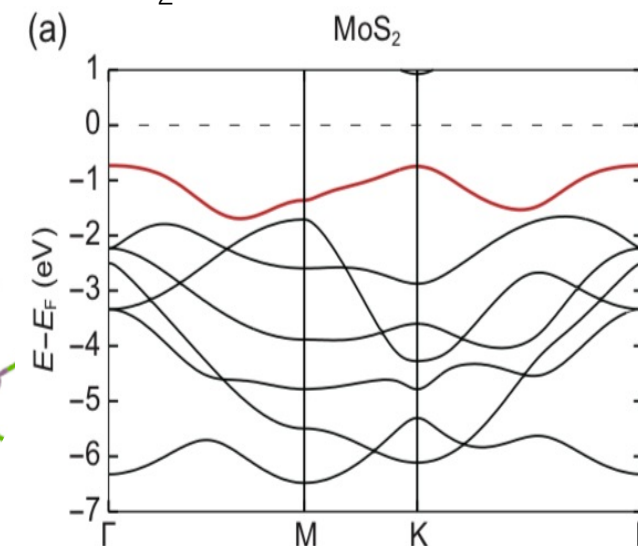
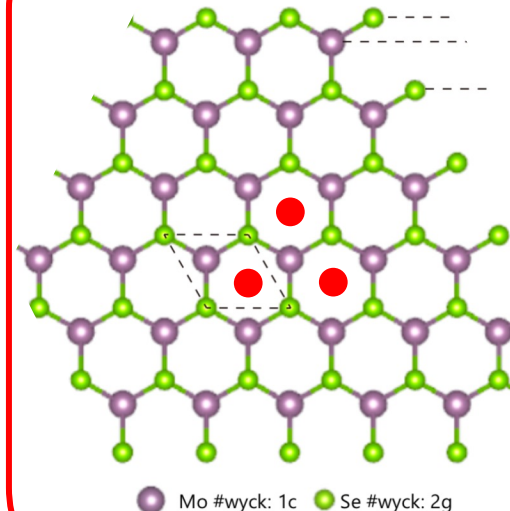
研究人员利用非常规材料Nb₃Br₈ 提出了一种无需外场的约瑟夫森二极管，研究结果发表在Nature 604, 653-656 (2022)。



EBR (A_{1g}@3b)

Without plotting the electronic distribution

1H MoS₂

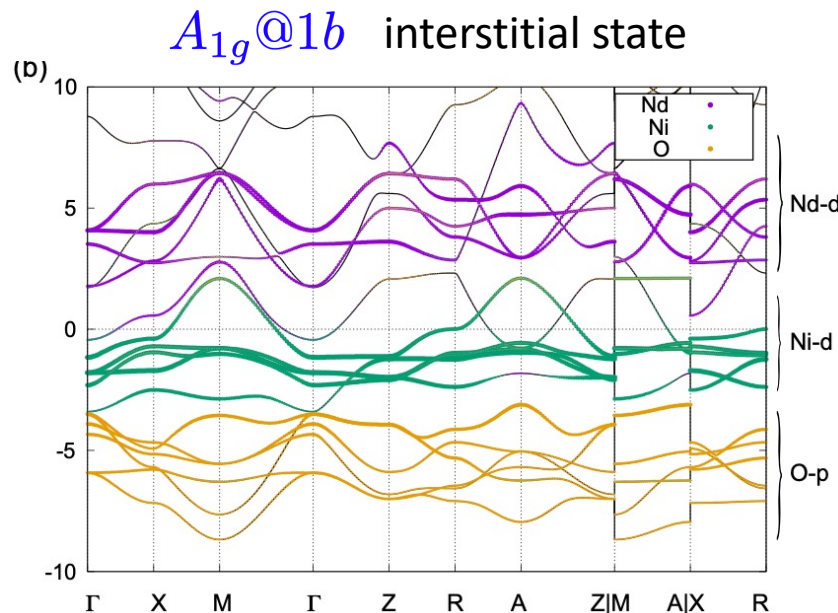
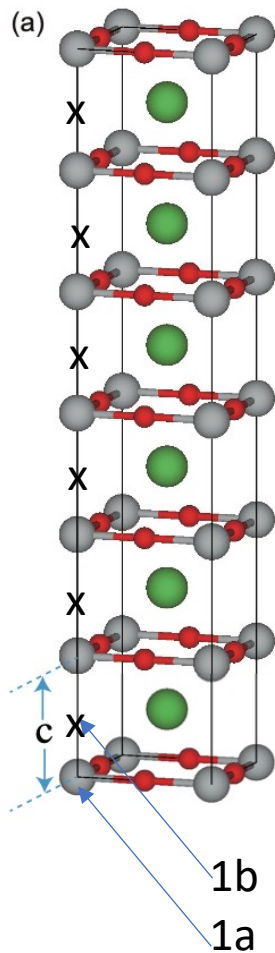


EBR (A_{1'}@1e)

Sci. Bull. 67, 598 (2022)

SPCMA 67, 246811 (2024)

镍基超导体NdNiO₂ with electrider-like state



These Fermi surfaces and symmetries can be reproduced by our two-band model, which consists of two elementary band representations (EBRs): $B_{1g}@1a + A_{1g}@1b$ (electrider-like state)

1. NSR, nwaa218 (2020)
2. Sci. Bull. 67, 598 (2022)
3. SPCMA 67, 246811 (2024)

TABLE S2: Atomic positions and valence states of $P4/mmm$ NdNiO₂.

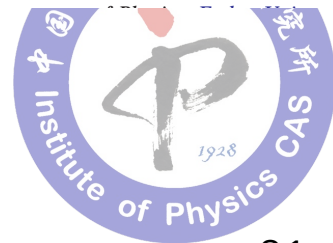
Atom	WKS(q)	Symm.	Irreps(ρ)	aBRs($\rho@q$)
Nd	1d	4/ mmm	$d_{z^2} : A_{1g}$	$A_{1g}@1d$
			$d_{x^2-y^2} : B_{1g}$	$B_{1g}@1d$
			$d_{xy} : B_{2g}$	$B_{2g}@1d$
			$d_{xz,yz} : E_g$	$E_g@1d$
Ni	1a	4/ mmm	$d_{z^2} : A_{1g}$	$A_{1g}@1a$
			$d_{x^2-y^2} : B_{1g}$	$B_{1g}@1a$
			$d_{xy} : B_{2g}$	$B_{2g}@1a$
			$d_{xz,yz} : E_g$	$E_g@1a$
O	2f	mmm	$p_z : B_{1u}$	$B_{1u}@2f$
			$p_y : B_{2u}$	$B_{2u}@2f$
			$p_x : B_{3u}$	$B_{3u}@2f$

PHYSICAL REVIEW LETTERS 135, 116501 (2025)

Editors' Suggestion

Observation of **Electriderlike** s States Coexisting with Correlated d Electrons in NdNiO₂

Chihao Li¹, Yutong Chen¹, Xiang Ding¹, Yezhao Zhuang², Nan Guo¹, Zhihui Chen¹, Yu Fan¹, Jiahao Ye¹, Zhitong An¹, Suppanut Sangphet¹, Shenglin Tang¹, Xiaoxiao Wang¹, Hai Huang², Haichao Xu^{1,3,*}, Donglai Feng^{4,†} and Rui Peng^{1,3,‡}



目录

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2. 拓扑能带表示理论（拓扑量子化学）
3. 能带表示的应用
 - a. 新型拓扑材料
 - b. 阻塞拓扑/非常规材料
4. 拓扑材料计算软件集 (TopMat)

Magnetic TQC

❑ 1651 magnetic space group (M=G+AG)

G is a unitary group, A is an antiunitary operator.

- ✓ Type I : 230 A = 0
- ✓ Type II : 230 A = time reversal (T)
- X Type III: 674 A = TR, with
R = not a pure translation
- X Type IV: 517 A = TR,
with R = pure translation

Co-irreps of a MSG (M=G+AG)

Herring Rules:

$$\frac{1}{|\mathbf{G}|} \sum_{B \in \mathbf{AG}} \chi_{\rho}(B^2) = \begin{cases} 1 & \text{case (a),} \\ -1 & \text{case (b),} \\ 0 & \text{case (c),} \end{cases}$$

$\rho_i^{(a)}$
 $\rho_i^{(b)} \rho_i^{(b)}$
 $\rho_i^{(c)} \rho_j^{(c)}$

- No eigenvalue for A; no additional irrep for M
- Degeneracy

❑ IRVSP (VASP) - <https://github.com/zjwang11/irvsp>

❑ IR2PW (QE) - <https://github.com/zjwang11/ir2pw>

How to get the co-irrep of a MSG?

1. identify its unitary subgroup G and get the irreps (ρ_i); 2. figure out the degeneracy of them.

MTQC: MSG = Unitary part (G) + Antiunitary part (AG)

- ❑ **Co-irreps at k:** magnetic k little group -> co-irreps -> vector data
- ❑ **Magnetic EBR (MEBR):** magnetic point groups -> co-irreps -> MBR -> MEBR
- ❑ **Compatibility Relations (CR):** CR of G -> CR of M
- ❑ magnetic symmetry indicators.

Ref : Gao, et al. PRB 106, 035150 (2022) + **Appendices (3000+ pages)**



磁性拓扑材料搜索

A generate workflow for topological materials in 1651 magnetic space groups (MSG)

Ref: Gao, J. et al. "Band representations in 1651 magnetic space groups", Phys. Rev. B 106, 035150 (2022).

1) prepare the original POSCAR file (without magnetism).

2) `phonopy --symmetry --tolerance 0.01 -c POSCAR`

3) `pos2msg` (converting PPOSCAR to POSCAR_msg and initializing MAGMOM on magnetic atoms)

#SG (1~230): #MSG (1~1651): (OG setting)

(* paste PPOSCAR below *)

`pos2msg`

4) `cp POSCAR_msg POSCAR`; do spin-polarized VASP calculations with the given "MAGMOM" setting.

5) `irvsp -sg sg#B [-nb $m $n] > outir` (generating tqc.txt/tqc.data when using only maximal HSKPs)

Note: sg#B is the corresponding space group number for unitary part of the MSG (found in the first line of POSCAR_msg).

6) solve compatibility relations (CR) and calculate symmetry indicators (SI) in 1651 magnetic space groups (using tqc.data).

#MSG (1~1651): (OG setting)

(* paste tqc.data below *)

Note: valid for 1651 magnetic space groups with spin-orbit coupling !

`solve_CR` (if not, it is a symmetry enforced semimetal)

`cal_SI`

http://tm.iphy.ac.cn/TopMat_1651msg.html

2025全国群论研讨会

能带理论和拓扑材料计算 TOPMAT.org

□ POS2MSG (Eu₃In₂As₄, 预测磁构型)

OG setting	BNS setting	MSG type	Detailed Inf.	integer spin	half-integer spin
58.1.471 <i>Pnnm</i>	<i>Pnnm</i> #58.393	I	Table MSG471	Z_2	Z_2
58.2.472 <i>Pnnm1'</i>	<i>Pnnm1'</i> #58.394	II	Table MSG472	Z_2	Z_4
58.3.473 <i>Pn'nm</i>	<i>Pn'nm</i> #58.395	III	Table MSG473	$\emptyset$	$\emptyset$
58.4.474 <i>Pnnm'</i>	<i>Pnnm'</i> #58.396	III	Table MSG474	$\emptyset$	$\emptyset$
58.5.475 <i>Pn'n'm</i>	<i>Pn'n'm</i> #58.397	III	Table MSG475	$Z_2 \times Z_2$	$Z_2 \times Z_2$
58.6.476 <i>Pnn'm'</i>	<i>Pnn'm'</i> #58.398	III	Table MSG476	Z_2	Z_2
58.7.477 <i>Pn'n'm'</i>	<i>Pn'n'm'</i> #58.399	III	Table MSG477	$\emptyset$	$\emptyset$

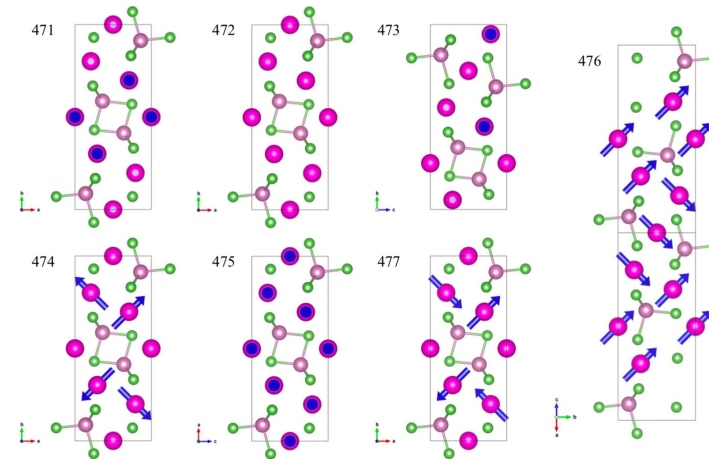
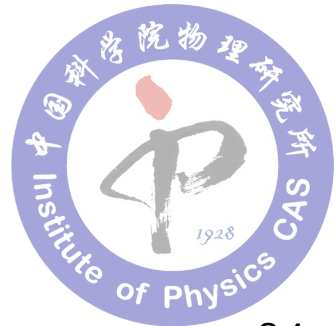


FIG. 1: The magnetic configurations of Eu₃In₂As₄. Numbers 471-477 is the MSG number (of OG setting).

□ MOM2MSG (已知磁构型)



材料计算软件集TopMat

	软件名	计算量	用途	备注
2010	parity	中心反演	有中心反演的拓扑绝缘体	VASP, WIEN2k
2012	IRVSP_v1	所有对称性操作的 trace	简单空间群k点不可约表示	点群特征标表
2014	IRVSP_v2 IR2TB		非简单空间群k点不可约表示 (时间, 用处)	空间群的特征标表 (BCS)
2018	CheckMatTop	对称性指标	晶体拓扑材料	230空间群的能带表示理论 TQC
2021	ABR/EBR decomposition		非常规材料 (阻塞拓扑, 电子盐, 超导)	
2022	TopMat 1651MSG	磁性对称性指标	磁性拓扑材料	1651磁性能带表示理论 MTQC
2023	VASP2KP	操作的表示矩阵	材料的低能有效模型 (不用拟合模型参数)	
2025	IRSSG		自旋群k点的不可约表示	自旋群的特征标表 (HSP)

TopMat软件集: topmat.org

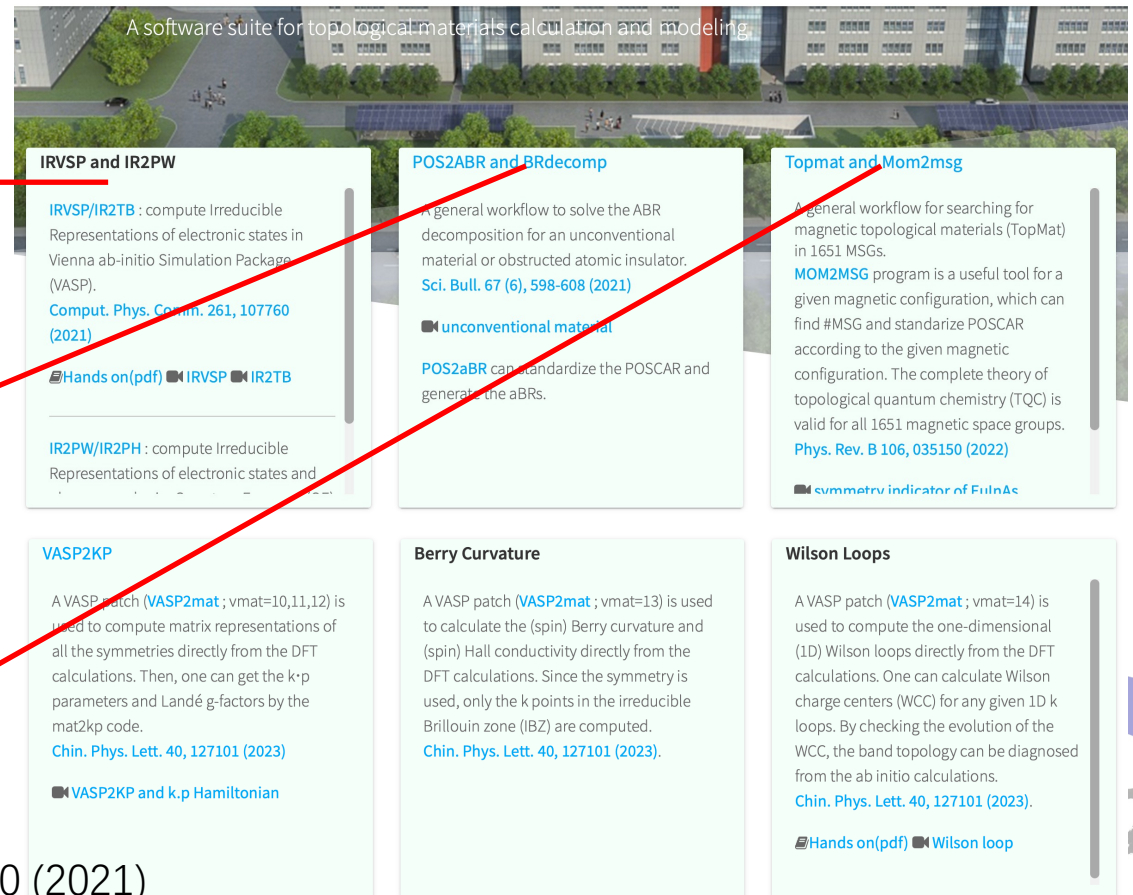
- Basis - irreps
the presentations of k little groups for 230 SGs
第一性计算得到**能带电子、声子的表示和共表示**
 - ✓ VASP: IRVSP (IRVSP library)
 - ✓ Phonon/Wannier TB: IR2TB
 - ✓ Quantum Espresso: IR2PW
 - ✓ Phonon modes: IR2PH
- Band Representation decomposition (BRdecomp)
——能带表示的分解 (Unconvention; OAI)
得到**非常规材料体系**
- Compatibility Relations (CR, 相容性关系)
——拓扑分类、对称性指标
得到**(磁性) 拓扑材料**

1. Gao et al. Computer Physics Communications 261, 107760 (2021)
2. Zhang et al. Chin. Phys. Lett. 40, 127101 (2023)

拓扑材料计算软件集

TopMat

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TopMat软件集 : topmat.org

基于DFT的波函数进行计算

VASP2mat功能

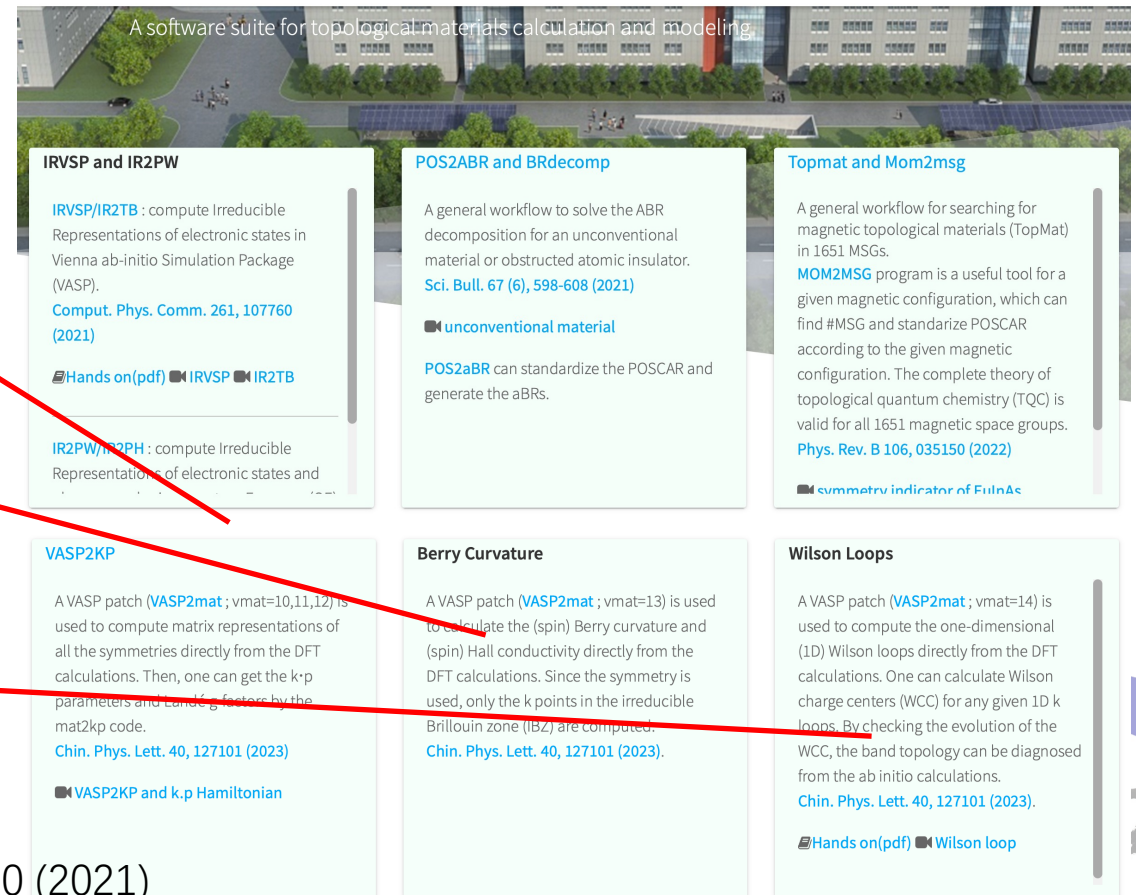
- ❑ vmat=10,11,12 与mat2kp结合, 计算材料的**kp模型, 参数和g factor**。
- ❑ vmat=13 与 symmat (github下载) 结合, 计算材料的**(自旋) 贝里曲率和 (自旋) Hall电导**。
- ❑ vmat=14 计算材料的Wilson loop, **Berry phases和拓扑不变量**

1. Gao et al. Computer Physics Communications 261, 107760 (2021)
2. Zhang et al. Chin. Phys. Lett. 40, 127101 (2023)

拓扑材料计算软件集

TopMat

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基于wannier的拓扑物性研究：WannierTools



国内自主开源软件

谷歌学术引用: 2656

中国科学院物理研究所

吴泉生等

quansheng.wu@iphy.ac.cn

Comp. Phys. Comm. 224, 405 (2018)

<https://www.wanniertools.org>

https://github.com/quanshengwu/wannier_tools

适用体系： 三维晶体材料、二维材料、Moire体系、声子体系等

拓扑诊断：

Z2拓扑绝缘体、Chern绝缘体

Wilson loop, Berry相位,

Berry曲率、狄拉克半金属、

外尔半金属、节点线半金属等分类

拓扑超导（华科徐刚）

物性计算：

能带、费米面、能带解折叠

表面态（ARPES/STM）、

Slab能带、Ribbon能带

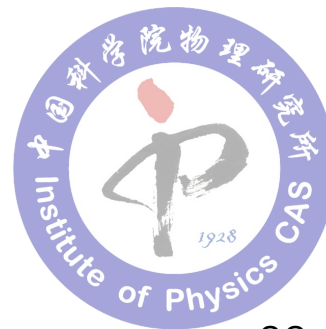
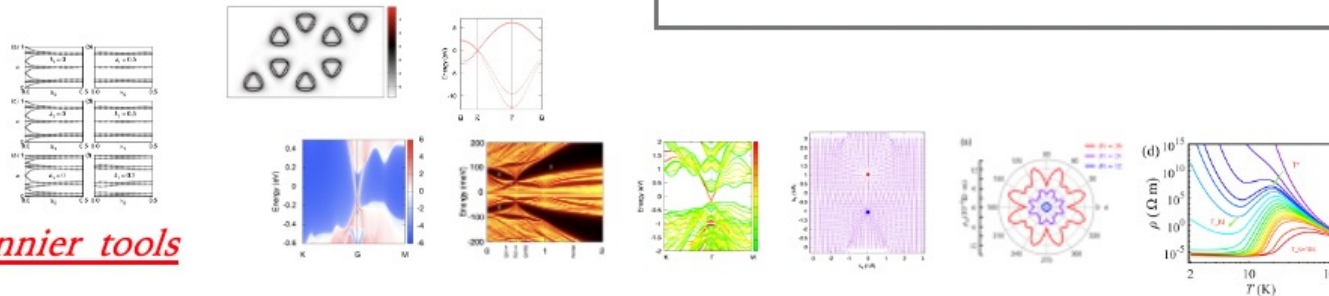
反常Hall效应、反常能斯特效应

自旋霍尔效应、轨道霍尔效应

光电导、非线性光学输运

Landau能级

国际唯一特色：非微扰计算磁电输运



Acknowledgement

- T03 group:



Profs. Zhong Fang(方忠)

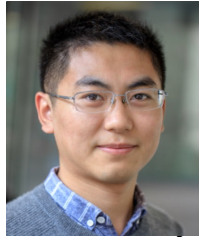


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Thanks for your attention!